

Simplex - Wolfe's Revised

$$\min Z = \frac{1}{2} w^T Q w + c^T w$$

$$\text{here } Z = 0.0064 w_1^2 + 0.0144 w_2^2 + 0.00576 w_1 w_2$$

$$\therefore Q = \begin{bmatrix} 0.0128 & 0.00576 \\ 0.00576 & 0.0288 \end{bmatrix} \quad c = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\text{s.t. } Aw = b, \quad A = \begin{bmatrix} 1 & 1 \end{bmatrix} \quad b = \begin{bmatrix} 1 \end{bmatrix}$$

$$\bullet \text{ KKT : } L(w, \lambda, \mu) = \frac{1}{2} w^T Q w + c^T w - \lambda^T (Aw - b) - \mu^T w$$

$$\nabla_w L = 0 \text{ gives,}$$

$$Qw + c - A^T \lambda - \mu = 0$$

$$\therefore \begin{bmatrix} 0.0128 & 0.00576 \\ 0.00576 & 0.0288 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \end{bmatrix} \lambda - \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} = 0$$

$$\text{thus, } \bullet 0.0128 w_1 + 0.00576 w_2 = \lambda + \mu_1 \quad \text{--- (i)}$$

$$\bullet 0.00576 w_1 + 0.0288 w_2 = \lambda + \mu_2 \quad \text{--- (ii)}$$

$$\text{also, } \bullet w_1 + w_2 = 1 \quad \text{--- (iii)}$$

$$\bullet w_1, w_2, \mu_1, \mu_2 \geq 0$$

$$\bullet \mu^T w = \mu_1 w_1 + \mu_2 w_2 = 0$$

$$\Rightarrow \mu_1 w_1 = \mu_2 w_2 = 0$$

$$\lambda \equiv \text{unrestricted, } \therefore \lambda = \lambda_1 - \lambda_2; \quad \lambda_1, \lambda_2 \geq 0$$

$$\therefore \text{ constraints : } \bullet 0.0128 w_1 + 0.00576 w_2 - \lambda_1 + \lambda_2 - \mu_1 + v_1 = 0$$

$$\bullet 0.00576 w_1 + 0.0288 w_2 - \lambda_1 + \lambda_2 - \mu_2 + v_2 = 0$$

$$\bullet w_1 + w_2 + v_3 = 1$$

$$\text{where } w_1, w_2, \lambda_1, \lambda_2, \mu_1, \mu_2, v_1, v_2, v_3 \geq 0$$

$$\text{Objective : minimise } Z = v_1 + v_2 + v_3$$

$$\text{Modification to Simplex : CST : } \mu_i w_i = 0$$

if w_i is currently in basis, μ_i cannot be pivoted to basis

Q is Positive definite (quadratic) and

constraint is a hyperplane : $w_1 + w_2 = 1$

Consider $(w_1, w_2) = (0, 1)$

if we move to $(\delta, 1-\delta)$, $\delta \rightarrow 0$

then functional value decreases,

... similarly for $(1, 0)$

thus minimum occurs at $w_1, w_2 > 0$

$\therefore$ by CST, $\mu_1 = \mu_2 = 0$

$$\text{or } 0.0128 w_1 + 0.00576 w_2 = 0.00576 w_1 + 0.0288 w_2$$

$$\text{or } w_1 = \frac{36}{11} w_2$$

$$\hookrightarrow w_1 + w_2 = \frac{47}{11} w_2 = 1 \quad \text{or } w_2 = \frac{11}{47}, \quad w_1 = \frac{36}{47}$$

$$\sim 0.234 \quad \sim 0.766$$

Tableau

Basic	w_1	w_2	λ_1	λ_2	μ_1	μ_2	v_1	v_2	v_3	RHS	γ
v_1	0.0128	0.00576	-1	1	-1	0	1	0	0	0	0
$\leftarrow v_2$	0.00576	0.0288	-1	1	0	-1	0	1	0	0	0
v_3	1	1	0	0	0	0	0	0	1	1	0
Z-C	1.01856	1.03456	-2	2	-1	-1	0	0	0	1	
v_1	0.0116	0	-0.8	0.8	-1	0.2	1	-0.2	0	0	-
w_2	0.2	1	-34.72	34.72	0	-34.72	0	34.72	0	0	-
$\leftarrow v_3$	0.8	0	34.72	-34.72	0	34.72	0	-34.72	1	1	
Z-C	0.8116	0	33.92	-33.92	-1	34.92	0	-35.92	1	1	
$\leftarrow v_1$	0.3	0	0	0	-1	1	1	-0.8	0.8	0.023	0.767
w_2	1.0	1	0	0	0	0	0	0	1	1.0	1.0
λ_1	0.230	0	1	-1	0	1	0	-1.0	0.028	0.028	1.252
Z-C	0.03	0	0	0	-1	1	0	0	0	0.023	
w_1	1	0	0	0	-33.24	33.24	33.24	-33.24	0.76	0.766	
w_2	0	1	0	0	33.24	-33.24	-33.24	33.24	0.234	0.234	
λ_1	0	0	1	-1	0.76	0.234	-0.76	-0.234	0.011	0.0112	
Z-C	0	0	0	0	0	0	-1	-1	-1	0	

$$\therefore Z_{\text{optimal}} = 0 = v_1 + v_2 + v_3$$

$$w_1 = 0.766, \quad w_2 = 0.234 \quad (\text{verified})$$